

NURUL NAZLEEN SUHAIRI

Senior Manager, Asset Integrity | Process Safety & Operational Excellence Leader

Brunei Darussalam | +60 11-1922 7031 | nurul.nazleen.suhairi@gmail.com | [linkedin.com/in/nurul-nazleen-suhairi/](https://www.linkedin.com/in/nurul-nazleen-suhairi/)

SUMMARY

Senior engineering leader with **19+ years of experience** across process safety, asset integrity, operational risk and technical leadership in complex oil & gas, petrochemical and fertilizer environments. Trusted advisor to senior leadership on critical engineering and risk decisions, with experience building organisational frameworks, strengthening operational performance and leading multidisciplinary teams across Malaysia and Brunei. Combines deep technical expertise with a strategic leadership approach, with a strong track record of turning complex risk and operational challenges into practical improvements in **safety, integrity, performance and organisational capability**.

CORE EXPERTISE

- **Leadership:** Executive Advisory, Strategic Leadership, Cross-Functional Leadership, Team Development, Coaching & Mentoring
- **Process Safety:** Process Safety Management, Process Hazard Analysis, HAZOP, ALARP, Loss of Primary Containment, Fire Prevention, Incident Management
- **Asset & Operations:** Asset Integrity, Asset Performance Management, Operational Excellence, Reliability, Maintenance Strategy, Operational Risk
- **Risk & Governance:** Risk Assessment, Risk Mitigation, Management of Change, Technical Governance, Engineering Assurance, Barrier Management
- **Transformation:** Digitalisation, Technology Integration, Continuous Improvement, Organisational Transformation, Performance Improvement
- **HSE & Compliance:** HSE Management, Environmental Compliance, Regulatory Compliance, ISO 9001, ISO 14001, ISO 45001, ISO 18001
- **Stakeholder Management:** Senior Stakeholder Engagement, Multidisciplinary Collaboration, Global Knowledge Sharing, Industry Engagement

PROFESSIONAL EXPERIENCE

BRUNEI FERTILIZER INDUSTRIES SDN BHD | Brunei

Senior Manager, Asset Integrity

June 2026 – Present

- Lead asset integrity priorities across a complex industrial operating environment, supporting safe, reliable and sustainable asset performance.
- Provide senior technical leadership on asset integrity, risk and engineering matters.
- Align asset performance initiatives with operational priorities, organisational objectives and long-term reliability.
- Lead and develop teams to strengthen technical capability, collaboration and performance.

Technical Manager

December 2025 – June 2026

- Developed a structured **Operational Excellence framework** spanning technical, maintenance, operations and asset performance priorities.
- Led development of an **Asset Performance Management system**, integrating risk management practices and technology.
- Served as **Technical Authority** for engineering changes, Management of Change, operational risk and asset integrity decisions.
- Supported Operations internal audit activities and alignment with **OSHMS and QEMS** requirements.
- Directed asset integrity improvement initiatives and facilitated knowledge sharing across multidisciplinary and multinational teams.
- Strengthened cross-functional collaboration to embed quality, integrity and operational excellence across the organisation.

PETRONAS | Malaysia

Manager, Process Safety Leadership & Culture

January 2023 – December 2025

- Led development and enhancement of the organisation's **Process Safety Culture framework**, integrating risk management and technology.
- Served as Process Safety **Subject Matter Expert** for critical decisions involving process, technology, design and operational risk.
- Directed enterprise process safety initiatives and facilitated knowledge sharing across global teams.
- Led the Process Safety Incident Database Steering Committee, strengthening organisational learning from incidents and best practices.
- Built cross-functional engagement through targeted training, workshops and leadership programmes.
- Mentored emerging leaders and technical professionals to strengthen long-term organisational capability.
- Represented PETRONAS at international conferences, technical forums and industry engagements.

Manager, Process Safety Risk Management

January 2019 – December 2022

- Led risk assessment and mitigation strategies across operational units, strengthening risk visibility and incident prevention.
- Implemented programmes focused on leadership accountability, barrier integrity and proactive risk management.
- Established fire prevention and mitigation frameworks that achieved **zero major fire incidents** over a multi-year period.
- Developed safety dashboards to improve risk identification, monitoring and incident management.
- Led the Process Safety Focused Enhancement Programme to strengthen management systems and leadership accountability.
- Designed and delivered programmes covering risk management, incident prevention and fire safety preparedness.

ASEAN BINTULU FERTILIZER SDN BHD | Malaysia

Executive, Environmental Management

August 2017 – December 2018

- Led environmental compliance and management initiatives across plant operations.
- Managed environmental assessments, scheduled waste and regulatory requirements.
- Supported sustainability and environmental performance improvements without compromising operational priorities.
- Collaborated with the Department of Environment to resolve regulatory compliance matters.

Executive, Process Safety

April 2010 – July 2017

- Led process safety activities across plant operations, covering hazard identification, risk assessment and incident prevention.
- Conducted process safety studies across ammonia plant operations and high-risk process activities.
- Applied root cause analysis to incidents and translated findings into practical operational improvements.
- Supported improvement of the organisation's HSE performance and process safety maturity.
- Delivered training and engagement initiatives to strengthen risk awareness and safety culture.

Executive, Process Engineering & Operations — Ammonia Plant

July 2007 – December 2008

- Supported process engineering and operational improvement initiatives across ammonia plant systems.
- Conducted equipment performance studies and supported optimisation of critical plant systems.
- Contributed to safe and reliable plant operations through engineering analysis and technical support.

PETRONAS GAS BERHAD | Malaysia

QA/QC Engineer — Joint Integrity Team

April 2007 – June 2007

- Supported quality assurance and joint integrity activities during gas plant start-up.
- Contributed to achieving **zero leaks following turnaround activities**.
- Coordinated quality control activities for critical equipment and plant systems.

SELECTED CAREER IMPACTS

- **Safety Performance:** Established fire prevention and mitigation frameworks contributing to **zero major fire incidents** over a multi-year period.
- **Asset Performance:** Developed an **Asset Performance Management System** integrating asset strategy, risk management and technology innovation to strengthen asset performance, whereby **Overall Equipment Effectiveness (OEE) improved by 12% within 6 months of implementation**.
- **Operational Excellence:** Developed a structured framework connecting technical, maintenance, operations and asset performance priorities.
- **Technical Governance:** Served as Technical Authority for critical engineering changes, Management of Change, operational risk and asset integrity decisions.
- **Process Safety Culture:** Led organisation-wide initiatives to strengthen safety culture, accountability, risk visibility and knowledge sharing; established **Organisational HSE Culture Maturity Measurement with benchmarking to the Global Energy Industry**.
- **Digital Risk Management:** Developed safety dashboards to improve visibility and management of process safety risks, providing risk visibility **from frontliners to the Executive Leadership Team and Board of Directors**.
- **Leadership Development:** Mentored emerging leaders and technical professionals, with **at least three mentees achieving certification as CIMAHS Competent Persons with DOSH Malaysia**, while multiple personnel progressed to **Staff Level through the Technical Professional Career Progression (TPCP)**.
- **Industry Influence:** Represented PETRONAS at international conferences, technical forums and professional engagements; served as **Co-Chairman for the Loss Prevention Asia Conference 2025**.

EDUCATION

Master of Process Safety and Loss Prevention

Universiti Putra Malaysia | 2019

Bachelor of Engineering — Chemical Engineering

The University of Queensland, Australia | 2006

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Major Hazards Competent Person — Department of Occupational Safety and Health (DOSH), Malaysia

Certified Environmental Professional in Scheduled Waste Management (CePSWaM) — Department of Environment (DOE), Malaysia

PETRONAS Certified HAZOP Leader — Level 1

NEBOSH International General Certificate in Occupational Health and Safety (IGC)

PETRONAS Technical Professional Career Progression (TPCP) Assessment — Staff Engineer, Process Safety

Qualified Auditor — ISO 9001, ISO 14001, ISO 45001 & ISO 18001

PROFESSIONAL AFFILIATION

Associate Member — Institution of Chemical Engineers (AMIChemE)

Graduate Engineer — Board of Engineers Malaysia

ADDITIONAL CREDENTIALS

Hazards Asia Pacific Symposium — Presenter

IChemE Regional Process Safety Seminar — Participant

PETRONAS Young Leaders Programme — Participant